

Confidence Intervals

ECON 97: Economic Toolkit

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<https://join.iclicker.com/RDNG>

Quiz: sample means

Let $\bar{X}$ be the sample mean of a random sample of size 9 from a distribution with pmf:

$$f(x) = 1/4 \quad x = 0, 1, 2, 3$$

What are $E[\bar{X}]$ and $\text{Var}(\bar{X})$?

- A. $3/18$ and $5/36$
- B. $3/2$ and $5/36$
- C. $3/18$ and $7/2$
- D. $3/2$ and $5/4$

Review of Concepts - Week 1

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Compliment: $P(A') = 1 - P(A)$
- Mutually exclusive: $P(A \cap B) = 0$
- Exhaustive: $P(A \cup B) = 1$
- $P(A \cup B)' = P(A' \cap B')$ and $P(A \cap B)' = P(A' \cup B')$
- $P(A \cap B') = P(A) - P(A \cap B)$
- Subset: if $A \subset B$ then $P(A \cap B) = P(A)$

Review of Concepts - Week 2

- Replacement, order matters: n^k
- No replacement
 - Order matters: ${}_nP_k = \frac{n!}{(n-k)!}$
 - Order doesn't matter: ${}_nC_k = \frac{n!}{(n-k)!k!}$
- ${}_nC_k = {}_nC_{(n-k)}$
- ${}_nC_n = 1$
- ${}_nC_1 = n$
- ${}_nC_0 = 1$

Review of Concepts - Week 3

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- $P(B|A) = \frac{P(A \cap B)}{P(A)}$
- If A, B independent ($A \perp B$):
 - $P(A \cap B) = P(A)P(B)$
 - $P(A|B) = P(A)$ and $P(B|A) = P(B)$
 - $A' \perp B'$, $A' \perp B$, and $A \perp B'$
 - $A \perp B \perp C \implies P(A \cap B \cap C) = P(A)P(B)P(C)$

Review of Concepts - Week 4

- **PMF**: $P(X = x) = f(x)$
 - E.g., $f(3) = P(X = 3)$
 - $\sum_{S_X} f(x) = 1$
 - Support of X (S_X): all possible values of X
- **CDF**: $P(X \leq x) = F(x)$
 - E.g., $F(2) = P(X \leq 2) = f(0) + f(1) + f(2)$
- **Mean**: $\mu = \sum_{S_X} x f(x)$
- **Variance**: $\sigma^2 = \sum_{S_X} (x - \mu)^2 f(x) = \sum_{S_X} x^2 f(x) - \mu^2$

Review of Concepts - Week 5

- Expectations

- $\mu = E[X] = \sum_{S_X} xf(x)$

- $E[g(X)] = \sum_{S_X} g(x)f(x)$

- E.g., $E[X^2] = \sum_{S_X} x^2 f(x)$

- $\sigma^2 = E[(X - \mu)^2] = E[X^2] - (E[X])^2$

- Properties

- $E[c] = c$

- $E[cX] = cE[X]$

- $E[X \pm Y] = E[X] \pm E[Y]$

- $E[aX \pm bY] = aE[X] \pm bE[Y]$

Review of Concepts - Week 6

Binomial: 2 outcomes; probability p and $(1 - p)$; n trials

$$X \sim \text{bin}(n, p), \quad x = 0, 1, \dots, n$$

$$f(x) = {}_n C_x (p)^x (1 - p)^{(n-x)}$$

$$E[X] = np \quad \text{Var}(X) = np(1 - p)$$

Poisson: Count number of events over time/space; average rate λ

$$X \sim \text{Pois}(\lambda), \quad x = 0, 1, 2, \dots$$

$$f(x) = (e^{-\lambda}) \frac{\lambda^x}{x!}$$

$$E[X] = \text{Var}(X) = \lambda$$

Review of Concepts - Week 7

Joint PMF: $f(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$

Marginal PMF: $f(x_1) = P(X_1 = x_1)$ and $f(x_2) = P(X_2 = x_2)$

→ if X_1 & X_2 independent, then $f(x_1, x_2) = f(x_1)f(x_2)$

Covariance: $E(X_1 X_2) - E(X_1)E(X_2)$

where $E(X_1 X_2) = \sum_{x_1} \sum_{x_2} (x_1 x_2) f(x_1, x_2)$

Correlation: $\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} \in [-1, 1]$

→ if X_1 & X_2 independent, then covariance and correlation are 0

Review of Concepts - Week 7

Consider random variables X_1, X_2

Means μ_1 and μ_2 ; variances σ_1^2 and σ_2^2 ; covariance σ_{12} ; correlation ρ_{12}

$$E[aX_1 \pm bX_2] = a\mu_1 \pm b\mu_2$$

$$\text{Var}(aX_1 \pm bX_2) = a^2\sigma_1^2 + b^2\sigma_2^2 \pm 2(ab)(\sigma_{12})$$

$$\sigma_{12} = \rho_{12}(\sigma_1\sigma_2)$$

If there are three variables:

$$\begin{aligned}\text{Var}(aX_1 \pm bX_2 \pm cX_3) &= a^2\sigma_1^2 + b^2\sigma_2^2 + c^2\sigma_3^2 \\ &\pm 2(ab)(\sigma_{12}) \pm 2(ac)(\sigma_{13}) \pm 2(bc)(\sigma_{23})\end{aligned}$$

Review of Concepts - Week 8

If $a \leq X \leq b$:

- $f(x)$: PDF of X
 - $\int_a^b f(x)dx = 1$
 - $P(X = a) = 0$
- $F(x) = P(X \leq x)$: CDF of X
 - $P(a \leq X \leq b) = \int_b^a f(x)dx = F(b) - F(a)$
 - $P(X \leq a) = P(X < a)$
 - $F'(x) = f(x)$
- $E[X] = \mu_X = \int_a^b xf(x)dx$
- $\text{Var}(X) = \sigma_X^2 = \int_a^b x^2 f(x)dx - \mu_X^2$

Review of Concepts - Week 8

Uniform: every outcome has equal probability

$$X \sim \text{Unif}(a, b), \quad a \leq x \leq b$$

$$f(x) = \frac{1}{b - a}$$

$$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases}$$

$$E[X] = \frac{a + b}{2} \quad \text{Var}(X) = \frac{(b - a)^2}{12}$$

Review of Concepts - Week 9

If a variable is normally distributed with $\mu = 0$ and $\sigma^2 = 1$

$$Z \sim N(0, 1)$$

$$F(x) = \Phi(x) = P(Z \leq x)$$

$$\Phi(-x) = 1 - \Phi(x)$$

$$\Phi(0) = 1/2$$

$$z_\alpha \implies \Phi(z_\alpha) = 1 - \alpha \text{ (percentile)}$$

If $X \sim N(\mu, \sigma^2)$:

$$\frac{X - \mu}{\sqrt{\sigma^2}} = Z \quad (\text{z-score})$$

Law of Large Numbers (LLN)

Chebyshev's Inequality:

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2} \quad \text{for any } k > 0$$

Take n “independent and identically distributed” (i.i.d.) observations with mean μ and variance σ^2

$$X_1, X_2, \dots, X_n$$

Then:

$$P(|\bar{X} - \mu| \geq \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

Interpretation: As sample size grows, the sample mean gets closer to the true mean

Intuition behind the LLN

Suppose the true mean height of UCLA students is 68 inches

If we take:

- A sample of 2 students, the average could vary a lot
- A sample of 200 students, the average is usually close to 68

Why?

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

Larger n means smaller variance for the sample mean

The Central Limit Theorem (CLT)

Let $\bar{X}$ be the sample mean from a random sample of size n .

If the population has mean μ and variance σ^2 , then:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

is approximately distributed as:

$$N(0, 1)$$

when n is sufficiently large (rule of thumb: $n \geq 30$ is enough)

Proof of CLT is complicated $\rightarrow$ easier to simulate (see file “clt_note.pdf” under R coding labs on course website)

What does the CLT mean?

Even if the original population is:

- skewed
- discrete
- non-normal

The distribution of the sample mean becomes approximately normal as $n \uparrow$

This is one of the most important ideas in statistics

It allows us to:

- build confidence intervals
- conduct hypothesis tests
- approximate probabilities

Standardizing the sample mean

To apply the CLT, we convert the sample mean into a standard normal variable:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

This measures:

- how far $\bar{X}$ is from μ
- in units of standard deviations

The standard “error” of the mean is:

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

CLT Example

Suppose:

- $n = 36$
- $\mu = 15$
- $\sigma = 2$

Approximate:

$$P(14.4 < \bar{X} < 15.6)$$

Standardize:

$$\begin{aligned} P\left(\frac{14.4 - 15}{2/\sqrt{36}} < Z < \frac{15.6 - 15}{2/\sqrt{36}}\right) \\ = P(-1.8 < Z < 1.8) \end{aligned}$$

CLT Example Continued

Using the standard normal table:

$$\begin{aligned}P(-1.8 < Z < 1.8) &= \Phi(1.8) - \Phi(-1.8) \\&= 0.9641 - 0.0359 \\&= 0.9282\end{aligned}$$

Therefore:

$$P(14.4 < \bar{X} < 15.6) \approx 0.9282$$

Another CLT Practice Problem

A random sample of size $n = 100$ is drawn from a population with:

$$\mu = 30, \quad \sigma = 3$$

Approximate:

$$P(\bar{X} > 29.5)$$

Key Takeaways

Law of Large Numbers

- The sample mean converges to the population mean

Central Limit Theorem

- The sample mean becomes approximately normal
- Works even when the original population is not normal

Important formulas

$$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

$$P(|\bar{X} - \mu| \geq k) \leq \frac{\sigma^2}{k^2}, \quad \forall k > 0$$

From the CLT to Confidence Intervals

The CLT tells us that for large n :

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1)$$

For a 95% confidence interval:

$$P(-1.96 < Z < 1.96) \approx 0.95$$

Substitute the standardized sample mean:

$$P\left(-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right) \approx 0.95$$

Solving for the Population Mean

Rearrange the inequality:

$$-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96$$

Multiply through by $\sigma/\sqrt{n}$:

$$-1.96 \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < 1.96 \frac{\sigma}{\sqrt{n}}$$

Solve for μ :

$$\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}$$

Confidence Interval Formula

In practice, we usually estimate σ using the sample standard deviation s .

General confidence interval form:

$$\text{estimate} \pm (\text{critical value}) \times (\text{standard error})$$

For a 95% confidence interval for the population mean:

$$\bar{X} \pm 1.96 \frac{s}{\sqrt{n}}$$

where

$$SE(\bar{X}) = \frac{s}{\sqrt{n}}$$

CLT + Confidence Interval practice